Lecture 6

Flows, symmetries and the symplectic form

Objectives. *Lecture 5 made the observables an algebra: every observable changes along a motion at the rate of its bracket with H. This lecture asks what flow an observable generates, and what that flow preserves. Every observable f generates a flow, as H generates the motion; momentum generates translations, angular momentum rotations. The flows preserve the bracket, and Jacobi is exactly the reason; brackets of observables become brackets of their flows; and a quantity is conserved exactly when H is invariant under its flow — Noether's theorem, now in both directions. Behind the bracket stands one bilinear form ω, the sum of oriented areas, and the flows preserve it: their Jacobian matrices are symplectic. For quadratic Hamiltonians this is the Lie algebra $\mathfrak{sp}(2n)$. With this, the second answer is almost complete.*

6.1 Hamiltonian vector fields: $X_f = J\nabla f$

As in Lecture 5, $\mathcal{M} \subseteq \mathbb{R}^{2n}$ is open, with coordinates $x = (q, p)$, and $J = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$ is the matrix of (4.9). Observables are C^∞ functions on $\mathcal{M}$ (Definition 5.1); in this lecture they do not depend on t.

Definition 6.1 (Hamiltonian vector field). The *Hamiltonian vector field* of $f \in C^\infty(\mathcal{M})$ is

$$X_f = J\nabla f = \Big(\frac{\partial f}{\partial p}, -\frac{\partial f}{\partial q}\Big).$$

Its flow φ_s^f (Theorem 1.5) is the *flow generated by* f; the parameter s need not be a time.

A vector field F on $\mathcal{M}$ acts on functions by differentiation along it, $Fh = \nabla h \cdot F$: along an integral curve γ, $\frac{\mathrm{d}}{\mathrm{d}s} h(\gamma(s)) = (Fh)(\gamma(s))$. For X_f this is a bracket:

$$X_f h = \nabla h \cdot J\nabla f = \sum_{i=1}^{n} \Big(\frac{\partial h}{\partial q_i}\frac{\partial f}{\partial p_i} - \frac{\partial h}{\partial p_i}\frac{\partial f}{\partial q_i} \Big) = \{h, f\}\,. \tag{6.1}$$

So along the flow of f, every observable h changes at the rate $\{h, f\}$, exactly as (5.2) says with f in place of H; Hamilton's equations (4.9) are $\dot{x} = X_H(x)$. Lecture 5 used the operator $D_f = \{f, \cdot\}$; by antisymmetry $D_f = -X_f$. We use X_f from now on.

Example 6.2 (Three generators). The flows below solve $\mathrm{d}x/\mathrm{d}s = X_f(x)$ with the stated initial value, as one checks by differentiating.

1. *Momentum generates translations.* $X_{p_i} = \partial/\partial q_i$, so $\varphi_s^{p_i}$ adds s to q_i and fixes the other coordinates.

2. *Angular momentum generates rotations.* On $\mathbb{R}^6$ let $L_z = q_1p_2 - q_2p_1$. By (6.1) and (5.10), $X_{L_z}u = \{u, L_z\} = -u \times e_z = e_z \times u$ for $u = q$ and $u = p$: the flow turns q and p together about e_z, counter-clockwise, through the angle s.
3. *A rotation of the phase plane* ($n = 1$). $f = \frac{1}{2}(q^2 + p^2)$ gives $X_f = (p, -q)$ and

$$\varphi_s^f(q, p) = (q\cos s + p\sin s,\ \ p\cos s - q\sin s). \tag{6.2}$$

Sign fact. The flow (6.2) is *clockwise* in the (q, p) plane: it carries $(1, 0)$ to $(\cos s, -\sin s)$ (Figure 6.1). The oscillator of Example 5.15 turns the same way, and the counter-clockwise rotation is generated by $-\frac{1}{2}(q^2 + p^2)$.

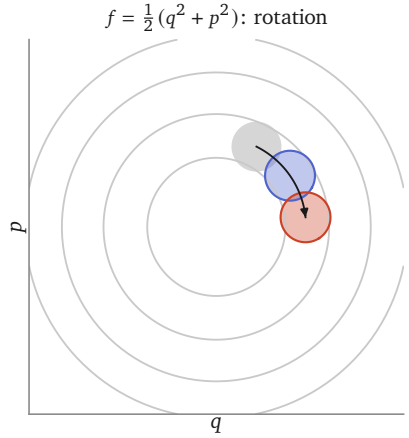


Figure 6.1: The flow of $f = \frac{1}{2}(q^2 + p^2)$: a disc of states (grey) and its images at $s = 0.5$ and $s = 1$, running clockwise along the level sets of f (faint).

Since $J^{\mathsf{T}} = -J$ and $J^{\mathsf{T}}J = I$,

$$\nabla H \cdot X_H = \nabla H^{\mathsf{T}} J \nabla H = 0, \qquad |X_H| = |\nabla H|\,.$$

At a point where $\nabla H \neq 0$, the gradient is normal to the level set through the point and X_H is tangent to it, with the same length. Conservation of energy is this geometry: $X_H H = \{H, H\} = 0$. The same holds for every observable: f is constant along its own flow.

6.2 Flows preserve the bracket, and Jacobi is why

In this section $H \in C^\infty(\mathcal{M})$ has a complete flow φ_t (Definition 1.8), and $f_t = f \circ \varphi_t$ is the Heisenberg evolution of Theorem 5.14: $\partial_t f_t = \{f_t, H\}$, and f_t is the only C^1 family with this property and $f_0 = f$.

Definition 6.3 (Poisson automorphism). A C^∞ bijection $\psi\colon \mathcal{M} \to \mathcal{M}$ with C^∞ inverse is a *Poisson automorphism* if $\{f \circ \psi, g \circ \psi\} = \{f, g\} \circ \psi$ for all $f, g \in C^\infty(\mathcal{M})$.

Theorem 6.4 (Theorem B: flows preserve the bracket · *proved modulo T2 (cited)*). *For all $f, g \in C^\infty(\mathcal{M})$ and all $t \in \mathbb{R}$,*

$$\{f, g\} \circ \varphi_t = \{f \circ \varphi_t, g \circ \varphi_t\}\,. \tag{6.3}$$

Each φ_t is a Poisson automorphism.

Proof. Let $h_t = \{f_t, g_t\}$. By Theorem 5.14(1), f_t and g_t are C^∞ in (t,x), so h is C^∞ and ∂_t commutes with the x-derivatives in the bracket:

$$\partial_t h_t = \{\partial_t f_t, g_t\} + \{f_t, \partial_t g_t\} = \{\{f_t, H\}, g_t\} + \{f_t, \{g_t, H\}\} = \{\{f_t, g_t\}, H\} = \{h_t, H\},$$

where the third equality is the Jacobi identity in the form (5.7). Also $h_0 = \{f, g\}$. By the uniqueness part of Theorem 5.14, $h_t = \{f, g\} \circ \varphi_t$. Each φ_t is a C^∞ bijection with inverse φ_{-t} (Theorem 1.5). ∎

Evolve-then-bracket equals bracket-then-evolve. Nothing in the proof uses that H is the energy: *if X_f is complete, every φ_s^f is a Poisson automorphism*, so the translations and rotations of Example 6.2 preserve the bracket. (For incomplete flows the same argument, run on the domain W of Theorem 1.5, gives (6.3) wherever φ_t is defined.)

For $n = 1$ the bracket of two functions is the Jacobian determinant of the pair: $\{Q, P\} = \det \partial(Q,P)/\partial(q,p)$. With $f = q$ and $g = p$, (6.3) reads $\det D\varphi_t = \{q_t, p_t\} = 1$: *a one-degree-of-freedom Hamiltonian flow preserves area* (Figure 6.2). § 6.6 and Lecture 7 extend this to every n.

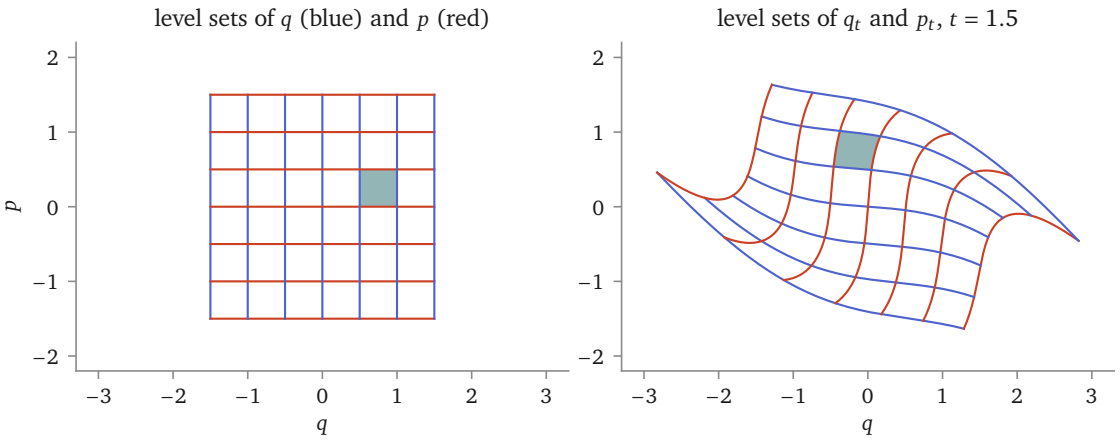


Figure 6.2: Theorem B for $f = q$, $g = p$. Left: the grid of level sets of q and p. Right: the level sets of q_t and p_t for the pendulum, $t = 1.5$, which are the grid lines moved by φ_{-t}. Since $\{q_t, p_t\} = \{q, p\} \circ \varphi_t = 1$, the shaded cell keeps its area $\frac{1}{4}$ (computed from the moved boundary).

The same argument works for other brackets, and it can be reversed. Let $U \subseteq \mathbb{R}^m$ be open, let $\Pi(x)$ be a C^∞ antisymmetric $m \times m$ matrix on U, and let $\{f, g\}_\Pi = \nabla f^{\mathsf{T}} \Pi \nabla g$ (Remark 5.6): bilinear, antisymmetric and Leibniz, as in Proposition 5.3; $m = 2n$ and $\Pi = J$ is our bracket, and nothing below uses that m is even. H evolves observables along the flow ψ_t of $\Pi\nabla H$, since $\nabla f \cdot \Pi \nabla H = \{f, H\}_\Pi$.

Theorem 6.5 (Jacobi ⇔ flows preserve the bracket · *proved modulo T1, T2 and smooth cut-offs (cited)*)**.** *$\{\cdot,\cdot\}_\Pi$ satisfies the Jacobi identity if and only if, for every $H \in C^\infty(U)$ with $\Pi\nabla H$ complete, every map ψ_t of its flow satisfies $\{f \circ \psi_t, g \circ \psi_t\}_\Pi = \{f, g\}_\Pi \circ \psi_t$ for all f, g.*

Proof. ($\Rightarrow$) The proofs of Theorem 5.14 and Theorem 6.4 use the bracket only through $\frac{\mathrm{d}}{\mathrm{d}t} f(\psi_t x) = \{f, H\}_\Pi(\psi_t x)$, through antisymmetry and through Jacobi; they hold verbatim for $\{\cdot,\cdot\}_\Pi$.

($\Leftarrow$) Fix f, g, H and $x_0 \in U$. Choose closed balls $B \subset K \subset U$ centred at x_0, and a C^∞ function χ equal to 1 on B and 0 outside K (a smooth cut-off; Lee, *Introduction to Smooth Manifolds*, Ch. 2). The field $F = \Pi\nabla(\chi H)$ vanishes outside K, so it is complete: a solution through a zero of F is constant (Theorem 1.3), so a solution starting in K never reaches a point outside K, and the blow-up alternative applies. Differentiate the hypothesis for χH at $t = 0$; the flow is C^∞ in (t, x) (Theorem 1.5), so ∂_t passes through the gradients, and at every point

$$\{\{f, \chi H\}_\Pi, g\}_\Pi + \{f, \{g, \chi H\}_\Pi\}_\Pi = \{\{f, g\}_\Pi, \chi H\}_\Pi .$$

At x_0 both sides involve derivatives of χH of order at most two, and $\chi H = H$ on B. So the identity holds at x_0 with H in place of χH. It is the second form of (5.7), which is (5.6) by antisymmetry. ∎

The meaning of Jacobi. The Jacobi identity is the statement that time evolution preserves the bracket, infinitesimally.

6.3 The Lie bracket, and $X_{\{f,g\}}$

A vector field $X = (a_1, \dots, a_m)$ is identified with the first-order operator $X = \sum_k a_k \partial_k$; its components are recovered as $a_k = X x_k$. So X_f is both the vector $J\nabla f$ and the operator $h \mapsto \{h, f\}$.

Definition 6.6 (Lie bracket)**.** The *Lie bracket* of vector fields $X = \sum a_k \partial_k$ and $Y = \sum b_k \partial_k$ is the commutator $[X, Y] = XY - YX$. By Lemma 5.4 it is again a vector field, with components

$$[X, Y]_k = X b_k - Y a_k, \qquad \text{that is} \qquad [X, Y] = DY\, X - DX\, Y, \tag{6.4}$$

where DX is the Jacobian matrix of the components.

The bracket is bilinear and antisymmetric, and it satisfies the Jacobi identity, since commutators of operators do. The symbol $[\cdot, \cdot]$ also denotes the commutator of matrices. The two differ by a sign on linear fields: for $X(v) = Av$ and $Y(v) = Bv$, (6.4) gives $[X, Y]\,(v) = (BA - AB)v$, the *negative* of the matrix commutator $[A, B]$. § 6.7 uses this.

Theorem 6.7 (Hamiltonian fields and brackets)**.** *For all $f, g \in C^\infty(\mathcal{M})$,*

$$X_{\{f,g\}} = -\left[X_f, X_g\right]. \tag{6.5}$$

Proof. Apply both sides to an observable h. By (6.1),

$$\begin{aligned} \left[X_f, X_g\right] h &= X_f\{h, g\} - X_g\{h, f\} = \{\{h, g\}, f\} - \{\{h, f\}, g\} \\ &= \{f, \{g, h\}\} + \{g, \{h, f\}\} = -\{h, \{f, g\}\}, \end{aligned}$$

by antisymmetry and then the Jacobi identity (5.6). The right side is $-X_{\{f,g\}} h$. Two first-order operators that agree on every h (in particular on the coordinates) are equal. ∎

The test case. In one degree of freedom take $f = q^2/2$ and $g = p^2/2$. Then $X_f = -q\,\partial_p$ and $X_g = p\,\partial_q$, and (6.4) gives

$$\left[X_f, X_g\right] = -q\,\partial_q + p\,\partial_p, \qquad \{f, g\} = qp, \qquad X_{qp} = q\,\partial_q - p\,\partial_p, \tag{6.6}$$

so $[X_f, X_g] = -X_{\{f,g\}}$.

The sign is the collection's convention (Bible §1). The map $f \mapsto X_f$ is linear and *reverses* brackets; $f \mapsto D_f = -X_f$ preserves them, $D_{\{f,g\}} = [D_f, D_g]$, which is the first form of (5.7).

Corollary 6.8 (Hamiltonian fields form a Lie algebra)**.** *The Hamiltonian vector fields on $\mathcal{M}$ form a Lie algebra under $[\cdot,\cdot]$. $X_f = 0$ if and only if $\nabla f = 0$, that is, f is locally constant. If $\mathcal{M}$ is connected, two observables generate the same field if and only if they differ by a constant.*

Proof. Closure is (6.5). J is invertible, so $J\nabla f = 0$ exactly when $\nabla f = 0$. Apply this to $f - g$. ∎

6.4 Noether reciprocity

Theorem 6.9 (Noether's theorem, Hamiltonian form · *proved; (ii)⇒(i) and (iii)⇒(i) use T1 (cited)*)**.** *For $H, J \in C^\infty(\mathcal{M})$ the following are equivalent:*

(i) $\{J, H\} = 0$*;*

(ii) *J is conserved: it is constant along every motion of H;*

(iii) *H is invariant under the flow of J: $H(\varphi_s^J x) = H(x)$ wherever $\varphi_s^J x$ is defined.*

Proof. (i)⇔(ii) is Proposition 5.10. For (iii), by (6.1) along the flow of J,

$$\frac{\mathrm{d}}{\mathrm{d}s} H(\varphi_s^J x) = \{H, J\}\,(\varphi_s^J x) = -\{J, H\}\,(\varphi_s^J x).$$

If (i) holds, the derivative vanishes on the whole interval of definition, which contains 0, so (iii) holds. If (iii) holds, the derivative vanishes at $s = 0$, so $\{J, H\}(x) = 0$; through every x passes an integral curve (Theorem 1.3). ∎

The statement is symmetric, since exchanging J and H in (i) changes only a sign: *J is conserved by the motion of H if and only if H is conserved by the flow of J.*

If the flow of J is complete, it also carries motions of H to motions: each $\psi = \varphi_s^J$ is a Poisson automorphism (Theorem 6.4) with $H \circ \psi = H$ by (iii), so $X_H(h \circ \psi) = \{h \circ \psi, H \circ \psi\} = (X_H h) \circ \psi$ for every h, and $\psi \circ x$ is a motion whenever x is. By Corollary 6.8, on a connected $\mathcal{M}$ the symmetry determines the conserved quantity up to a constant.

Point transformations: the cotangent lift

Which observables generate transformations of positions? Let $U \subseteq \mathbb{R}^n$ be open, $\mathcal{M} = U \times \mathbb{R}^n$, and let $\xi \in C^\infty(U, \mathbb{R}^n)$ be a complete vector field on U with flow h^s.

Proposition 6.10 (The cotangent lift · *proved modulo T1, T5 (cited)*)**.** *The flow of $J(q,p) = p \cdot \xi(q)$ is*

$$\varphi_s^J(q,p) = \big(h^s(q),\ Dh^s(q)^{-\mathsf{T}} p\big), \qquad s \in \mathbb{R}. \tag{6.7}$$

Proof. $X_J = (\partial J/\partial p, -\partial J/\partial q) = (\xi(q), -D\xi(q)^{\mathsf{T}} p)$. The q-equation $\mathrm{d}q/\mathrm{d}s = \xi(q)$ does not involve p, and is solved by $q(s) = h^s(q_0)$. Let $M(s) = Dh^s(q_0)$; by the variational equation of Theorem 1.7, $M' = D\xi(q(s))\,M$, $M(0) = I$. Then $p(s) = M^{-\mathsf{T}} p_0$ satisfies

$$p' = -M^{-\mathsf{T}} M'^{\mathsf{T}} M^{-\mathsf{T}} p_0 = -D\xi(q(s))^{\mathsf{T}} p(s),$$

the p-equation, with $p(0) = p_0$. By uniqueness (Theorem 1.3) this is the flow, and it is defined for all s. ∎

(6.7) is the covector law (4.3) applied to the point transformation h^s: positions move by h^s, and momenta move as covectors must (Figure 6.3). Translations ($\xi = e$, $J = e \cdot p$) and rotations ($\xi = e \times q$, $J = p \cdot (e \times q) = e \cdot \mathbf{L}$) are of this kind: these are the generators of Example 6.2.

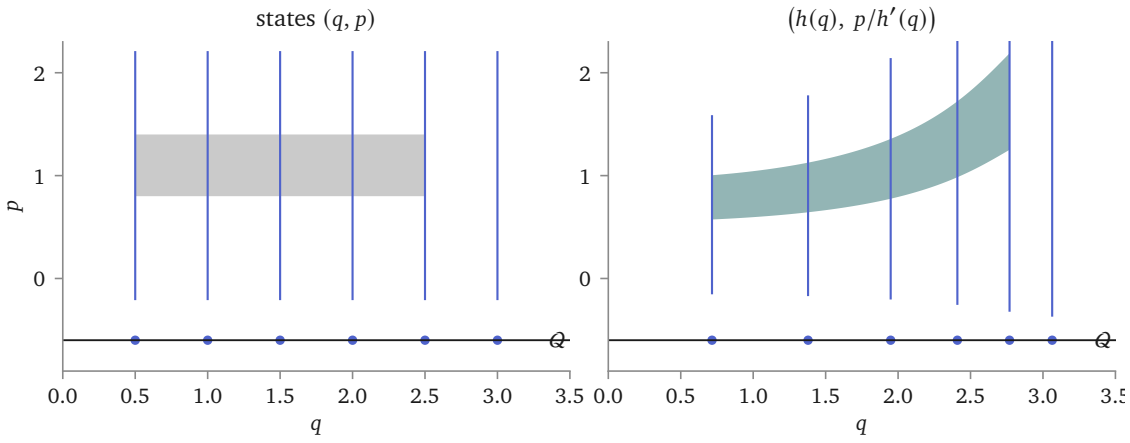


Figure 6.3: A lifted point transformation, $n = 1$: $(q, p) \mapsto (h(q), p/h'(q))$ with $h(q) = q + 0.45 \sin q$. Vertical lines go to vertical lines, and the grey rectangle keeps its area. Below: the configuration line and the points $h(q)$.

The charges of Theorem 3.13 are these $p \cdot \xi$. Theorem 6.9 admits every observable J; Lecture 10 meets a conserved one that is quadratic in p.

6.5 The symplectic form

The matrix J of (4.9) has $J^{\mathsf{T}} = -J = J^{-1}$, $J^2 = -I$ and $\det J = 1$.

Definition 6.11 (The symplectic form). The *symplectic form* on $\mathbb{R}^{2n}$ is

$$\omega(u, v) = u^{\mathsf{T}} J v = \sum_{i=1}^{n} \left(u_{q_i} v_{p_i} - u_{p_i} v_{q_i}\right), \tag{6.8}$$

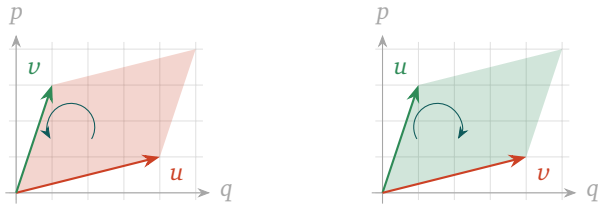
where $u = (u_q, u_p)$ and $v = (v_q, v_p)$.

It is bilinear; it is *antisymmetric*, $\omega(u, v) = -\omega(v, u)$, because $J^{\mathsf{T}} = -J$; and it is *non-degenerate*: if $\omega(u, v) = 0$ for every v then $J^{\mathsf{T}} u = 0$, and J is invertible, so $u = 0$. Antisymmetry gives $\omega(u, u) = 0$: every vector is ω-orthogonal to itself, which no inner product allows.

What it measures

For $n = 1$, (6.8) is $u_q v_p - u_p v_q = \det \begin{pmatrix} u & v \end{pmatrix}$, the *oriented area* of the parallelogram spanned by u and v, in that order: positive when v lies anticlockwise from u, and changing sign when the two are exchanged (Figure 6.4).

For $n > 1$, (6.8) is the *sum of the oriented areas of the projections of that parallelogram onto the n planes* (q_i, p_i); the projections onto (q_1, q_2), (q_1, p_2) and the other coordinate planes



$\omega(u,v) = +2.75$ $\omega(u,v) = -2.75$

Figure 6.4: One degree of freedom: $\omega(u,v)$ is the area of the parallelogram on u and v, signed by the sense in which the boundary u, v is traversed. The right panel is the same pair of vectors with the names exchanged. Here $u = (2, \frac{1}{2})$, $v = (\frac{1}{2}, \frac{3}{2})$ and $\omega(u,v) = 2 \cdot \frac{3}{2} - \frac{1}{2} \cdot \frac{1}{2}$.

do not enter. That is what the notation $\omega = \sum_i \mathrm{d}q_i \wedge \mathrm{d}p_i$ records: one oriented area for each pair (q_i, p_i).

Bracket, field and form are one structure

Proposition 6.12 (ω recovers the bracket). *For all $f, g \in C^\infty(\mathcal{M})$ and all $v \in \mathbb{R}^{2n}$,*

$$\omega(X_f, v) = \nabla f \cdot v, \qquad \text{and} \qquad \omega(X_f, X_g) = \{f, g\}\,. \tag{6.9}$$

Moreover X_f is the only vector field satisfying the first identity.

Proof. $\omega(X_f, v) = (J\nabla f)^{\mathsf{T}} J v = \nabla f^{\mathsf{T}} J^{\mathsf{T}} J v = \nabla f \cdot v$, since $J^{\mathsf{T}} J = I$. Putting $v = X_g = J\nabla g$ gives $\nabla f^{\mathsf{T}} J \nabla g$, which is $\{f, g\}$ by (5.1). If $\omega(w, v) = \nabla f \cdot v$ for all v, then $\omega(w - X_f, v) = 0$ for all v, so $w = X_f$ by non-degeneracy. ∎

So bracket, field and form are one structure, and of the three ω is the one that differentiates nothing.

6.6 The symplectic group; flows are symplectic

Definition 6.13 (Symplectic matrix). A real $2n \times 2n$ matrix A is *symplectic* if $A^{\mathsf{T}} J A = J$, equivalently if $\omega(Au, Av) = \omega(u, v)$ for all u, v. They form the set $\mathrm{Sp}(2n)$.

Proposition 6.14 ($\mathrm{Sp}(2n)$ is a group). *If $A \in \mathrm{Sp}(2n)$ then $\det A = \pm 1$; A is invertible with $A^{-1} = -JA^{\mathsf{T}}J \in \mathrm{Sp}(2n)$; and $AB \in \mathrm{Sp}(2n)$ for $B \in \mathrm{Sp}(2n)$. For $n = 1$, $A^{\mathsf{T}} J A = (\det A) J$, so $\mathrm{Sp}(2) = \mathrm{SL}(2, \mathbb{R})$.*

Proof. Taking determinants in $A^{\mathsf{T}} J A = J$ gives $(\det A)^2 \det J = \det J$ and $\det J = 1$, so $(\det A)^2 = 1$. In particular A is invertible. Multiplying $A^{\mathsf{T}} J A = J$ on the left by $-J$ and using $J^2 = -I$ gives $(-JA^{\mathsf{T}}J)A = -J^2 = I$, and a left inverse of an invertible matrix is its inverse. Then $(A^{-1})^{\mathsf{T}} J A^{-1} = (A^{-1})^{\mathsf{T}} (A^{\mathsf{T}} J A) A^{-1} = J$, and $(AB)^{\mathsf{T}} J (AB) = B^{\mathsf{T}} J B = J$. For $n = 1$ expand the 2×2 product. ∎

Symplectic is stronger than volume-preserving

For $n = 1$, Proposition 6.14 says that symplectic means $\det A = 1$: area and orientation preserved, nothing more. For $n \geq 2$ it is strictly stronger. With e_{q_i}, e_{p_i} the standard basis vectors, take, in the order (q_1, q_2, p_1, p_2),

$$A = \operatorname{diag}\left(2, 1, 1, \tfrac{1}{2}\right), \qquad \det A = 1. \tag{6.10}$$

Then $\omega(Ae_{q_1}, Ae_{p_1}) = 2$ and $\omega(Ae_{q_2}, Ae_{p_2}) = \frac{1}{2}$, instead of 1 and 1: the map doubles one projected area and halves the other. The two errors cancel in the determinant and not in ω.

Theorem 6.15 (Hamiltonian flows are symplectic · *proved modulo T2, T5 (cited)*)**.** *Let $H \in C^\infty(\mathcal{M})$ with flow φ_t. Then $D\varphi_t(x) \in \operatorname{Sp}(2n)$ for every x and every t for which $\varphi_t x$ is defined.*

Proof. Fix x and let $M(t) = D\varphi_t(x)$, which is C^1 in t on the open interval $I_x \ni 0$ where the motion exists (Theorem 1.5). Differentiating $\partial_t \varphi_t x = X_H(\varphi_t x)$ in x, the variational equation of Theorem 1.7 reads

$$M'(t) = J \operatorname{Hess} H(\varphi_t x)\, M(t), \qquad M(0) = I, \tag{6.11}$$

since $DX_H = D(J\nabla H) = J \operatorname{Hess} H$. Write $S = \operatorname{Hess} H(\varphi_t x)$, which is *symmetric*. Then

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(M^{\mathsf{T}} J M\right) = M^{\mathsf{T}}\left[(JS)^{\mathsf{T}} J + J(JS)\right]M = M^{\mathsf{T}}\left[SJ^{\mathsf{T}}J + J^2 S\right]M = M^{\mathsf{T}}\left[S - S\right]M = 0.$$

So $M(t)^{\mathsf{T}} J M(t) = M(0)^{\mathsf{T}} J M(0) = J$ on I_x. ∎

Only two facts were used: that X_H is J times a gradient, and that a Hessian is symmetric. For $n = 1$ this is the area statement of § 6.2; Lecture 7 proves in every dimension that volume is preserved.

6.7 Quadratic Hamiltonians

For a symmetric $2n \times 2n$ matrix S put

$$H_S(x) = \tfrac{1}{2} x^{\mathsf{T}} S x, \qquad \text{so that} \qquad X_{H_S}(x) = J\nabla H_S = JSx.$$

Every homogeneous quadratic is an H_S for exactly one symmetric S, and the motion is the linear equation $\dot{x} = JSx$, solved by $x(t) = e^{tJS}x_0$ for all $t \in \mathbb{R}$.

Proposition 6.16 (Hamiltonian matrices)**.** *For a real $2n \times 2n$ matrix X the following are equivalent: (a) $X^{\mathsf{T}}J + JX = 0$; (b) $X = JS$ with $S = -JX$ symmetric; (c) $e^{tX} \in \operatorname{Sp}(2n)$ for all $t \in \mathbb{R}$. These matrices form $\mathfrak{sp}(2n)$, a linear space of dimension $n(2n+1)$.*

Proof. (b)⇒(a): $(JS)^{\mathsf{T}}J + J^2 S = SJ^{\mathsf{T}}J - S = 0$. (a)⇒(b): put $S = -JX$; then $S^{\mathsf{T}} = X^{\mathsf{T}}J = -JX = S$ and $JS = X$. (a)⇔(c): $Z(t) = e^{tX^{\mathsf{T}}} J e^{tX}$ has $Z' = e^{tX^{\mathsf{T}}}(X^{\mathsf{T}}J + JX)e^{tX}$ and $Z(0) = J$, so (a) gives $Z \equiv J$, and differentiating $Z \equiv J$ at $t = 0$ gives (a). Since $S \mapsto JS$ is a bijection, the dimension is that of the symmetric matrices. ∎

Theorem 6.17 (Quadratics and matrices)**.** *For symmetric S, T,*

$$\{H_S, H_T\} = H_{S'}, \qquad S' = SJT - TJS, \qquad JS' = [JS, JT]\,, \tag{6.12}$$

> *the commutator of matrices. Hence $\mathfrak{sp}(2n)$ is closed under $[\cdot,\cdot]$, and $JS \mapsto H_S$ is a linear isomorphism of $\mathfrak{sp}(2n)$ onto the homogeneous quadratics that carries the matrix commutator to the Poisson bracket.*

Proof. $\{H_S, H_T\} = (Sx)^{\mathsf{T}} J(Tx) = x^{\mathsf{T}} SJTx$. A quadratic form only sees the symmetric part, and $(SJT)^{\mathsf{T}} = -TJS$, so $x^{\mathsf{T}} SJTx = \frac{1}{2} x^{\mathsf{T}} S'x = H_{S'}$ with S' as stated; S' is symmetric. Multiplying S' by J, $JS' = JSJT - JTJS = [JS, JT]$. Closure of $\mathfrak{sp}(2n)$ follows, since S' is symmetric. ∎

Proposition 5.16 already gave the closure; the theorem names the Lie algebra. The signs agree with § 6.3: for linear fields $X(x) = Ax$ and $Y(x) = Bx$ the vector-field bracket $[X, Y]\,(x) = [B, A]\,x$ is the negative of the matrix commutator (Definition 6.6), so $X_{\{H_S, H_T\}} = [JS, JT]\,x = -\left[X_{H_S}, X_{H_T}\right]$, which is (6.5).

The oscillator is the example. For $H = \frac{1}{2}\nu(q^2 + p^2)$, $S = \nu I$ and $JS = \nu J$, whose exponential $e^{t\nu J}$ is the clockwise rotation through the angle νt of (6.2) and Figure 6.1: an element of $\mathrm{Sp}(2)$ for every t, as Proposition 6.16(c) requires.

Summary.

- *Handed on:* every observable generates a flow, $X_f = J\nabla f$ (Definition 6.1); complete flows preserve the bracket, and a bracket satisfies Jacobi exactly when its flows do (Theorems 6.4 and 6.5); $X_{\{f,g\}} = -\left[X_f, X_g\right]$ (Theorem 6.7); J is conserved exactly when H is invariant under the flow of J (Theorem 6.9), and point symmetries are cotangent lifts (Proposition 6.10).
- *The form behind the bracket:* $\omega(u, v) = u^{\mathsf{T}} Jv$, with $\omega(X_f, X_g) = \{f, g\}$ (Proposition 6.12); flows have symplectic Jacobians (Theorem 6.15); the quadratic Hamiltonians are $\mathfrak{sp}(2n)$ (Theorem 6.17).
- *Next:* Lecture 7 puts the states beside the observables: volume is preserved, densities are carried along, and the second answer is complete.

References

Marsden and Ratiu, *Introduction to Mechanics and Symmetry*, §2.7 and §13.1. Takhtajan, *Quantum Mechanics for Mathematicians*, §2.7. Arnold, *Mathematical Methods of Classical Mechanics*, §§39–41. Meyer and Offin, *Introduction to Hamiltonian Dynamical Systems and the N-Body Problem*, §§2.1–2.2 and p. 196.